

4MA1_2H_ch5_vector_transformation_geometry

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1. 4MA1/2HR_Summer_2025_Q18

OAB is a triangle.

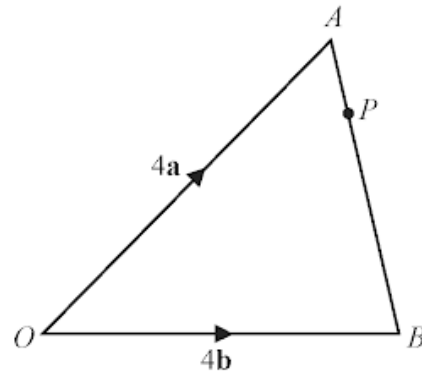


Diagram **NOT**
accurately drawn

$$\vec{OA} = 4\mathbf{a}$$

$$\vec{OB} = 4\mathbf{b}$$

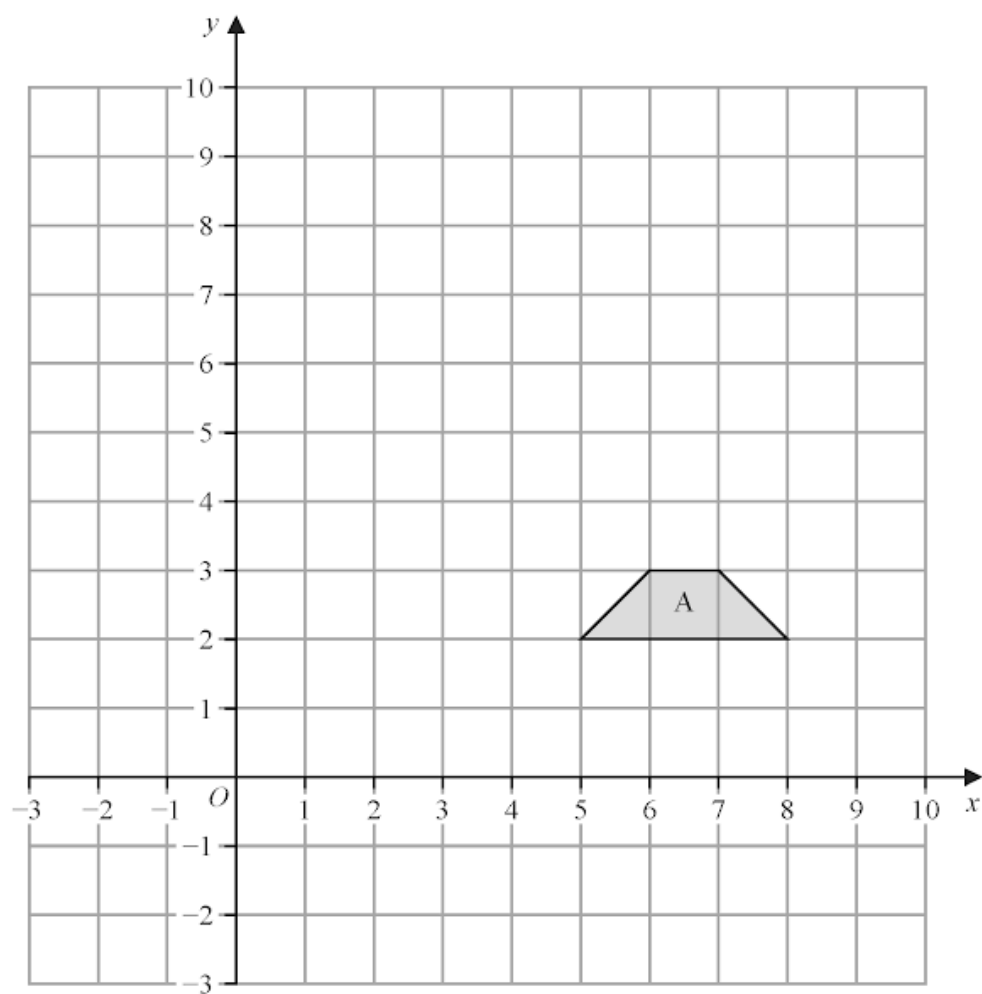
P is the point on AB such that $AP : PB = 1 : 3$

(a) Write down $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

.....
(1)

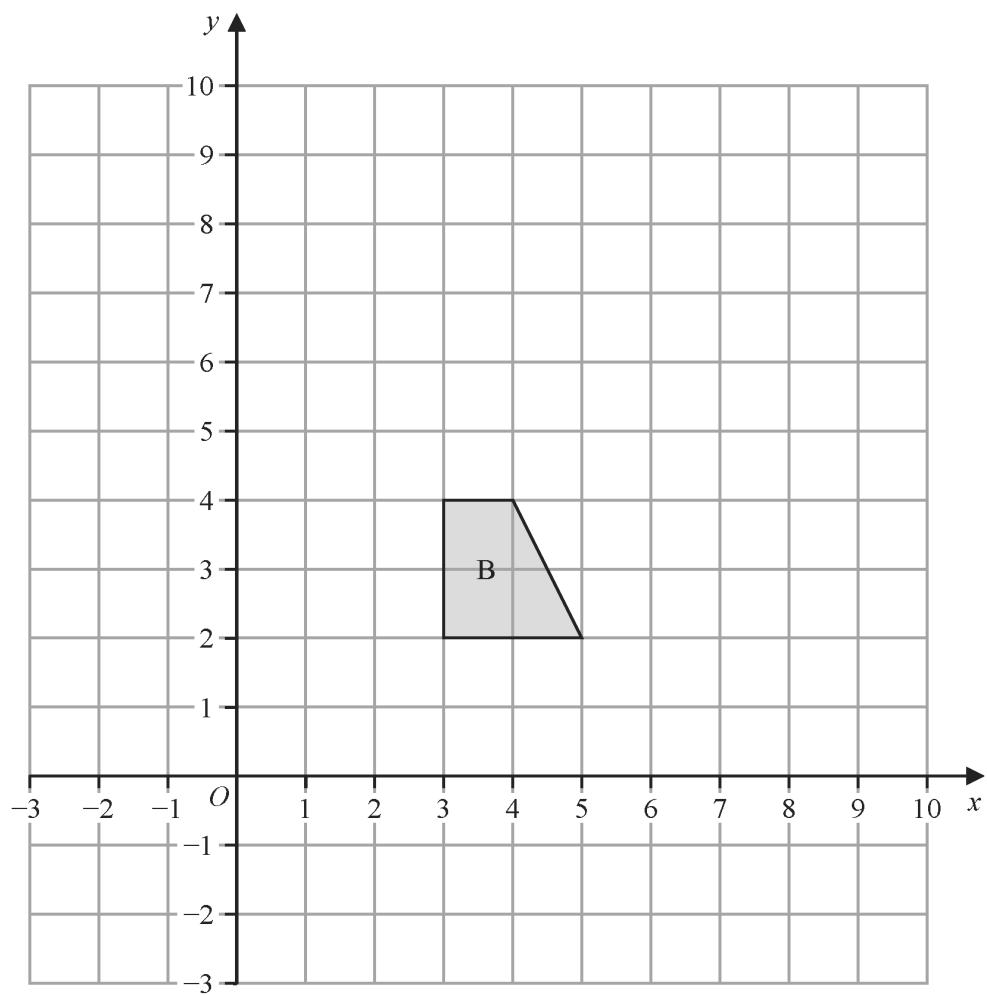
(b) Express $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

.....
(2)



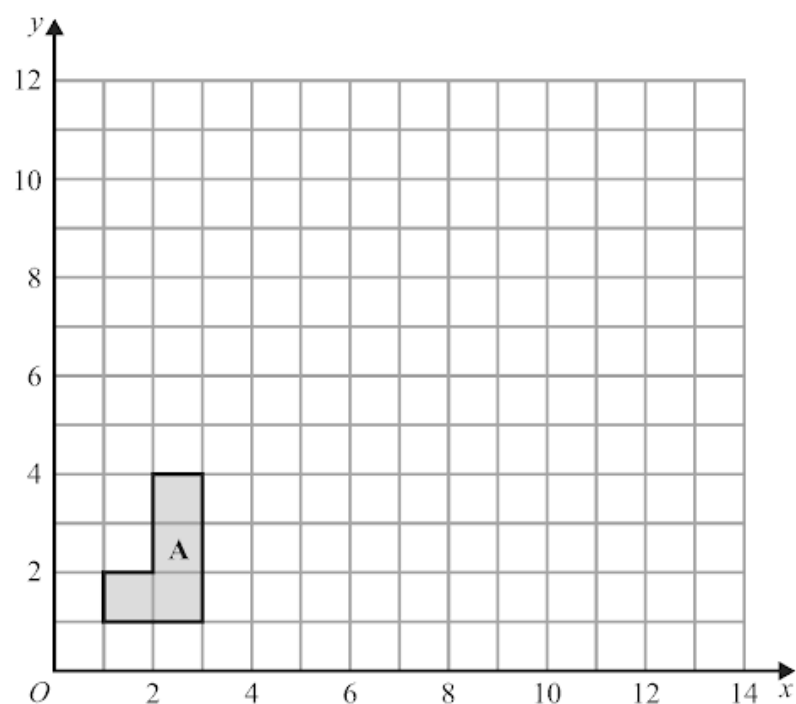
(a) On the grid above, reflect shape **A** in the line $y = x$

(2)



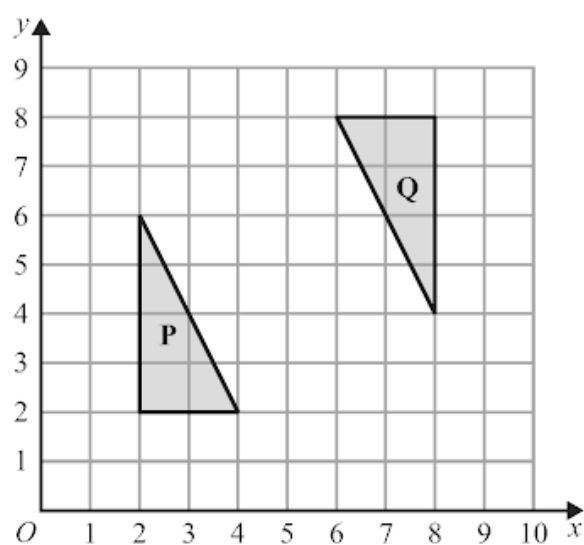
(b) On the grid above, enlarge shape **B** by scale factor 2 with centre (1, 1)

(2)



(a) On the grid, enlarge shape **A** with scale factor 3 and centre $(0, 1)$

(2)



(b) Describe fully the single transformation that maps triangle **P** onto triangle **Q**

(3)

4. 4MA1/2H_Winter_2024_Q17

Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

3 marks

$OPQR$ is a parallelogram.

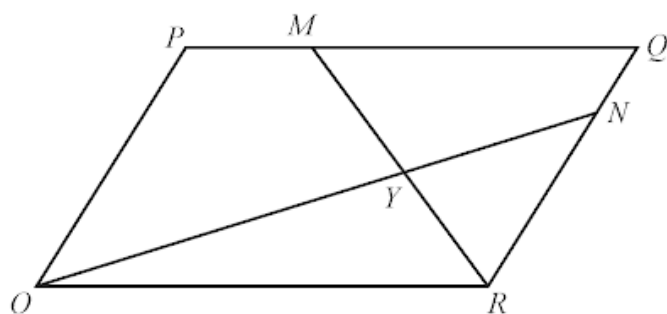


Diagram **NOT**
accurately drawn

$$\vec{OP} = 2\mathbf{a} \quad \text{and} \quad \vec{OR} = 3\mathbf{b}$$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answers in simplest form

(i) $\vec{ON}$

(1)

(ii) $\vec{MR}$

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

(4)

6. 4MA1/2HR_Summer_2024_Q21

$ABCD$ is a square.

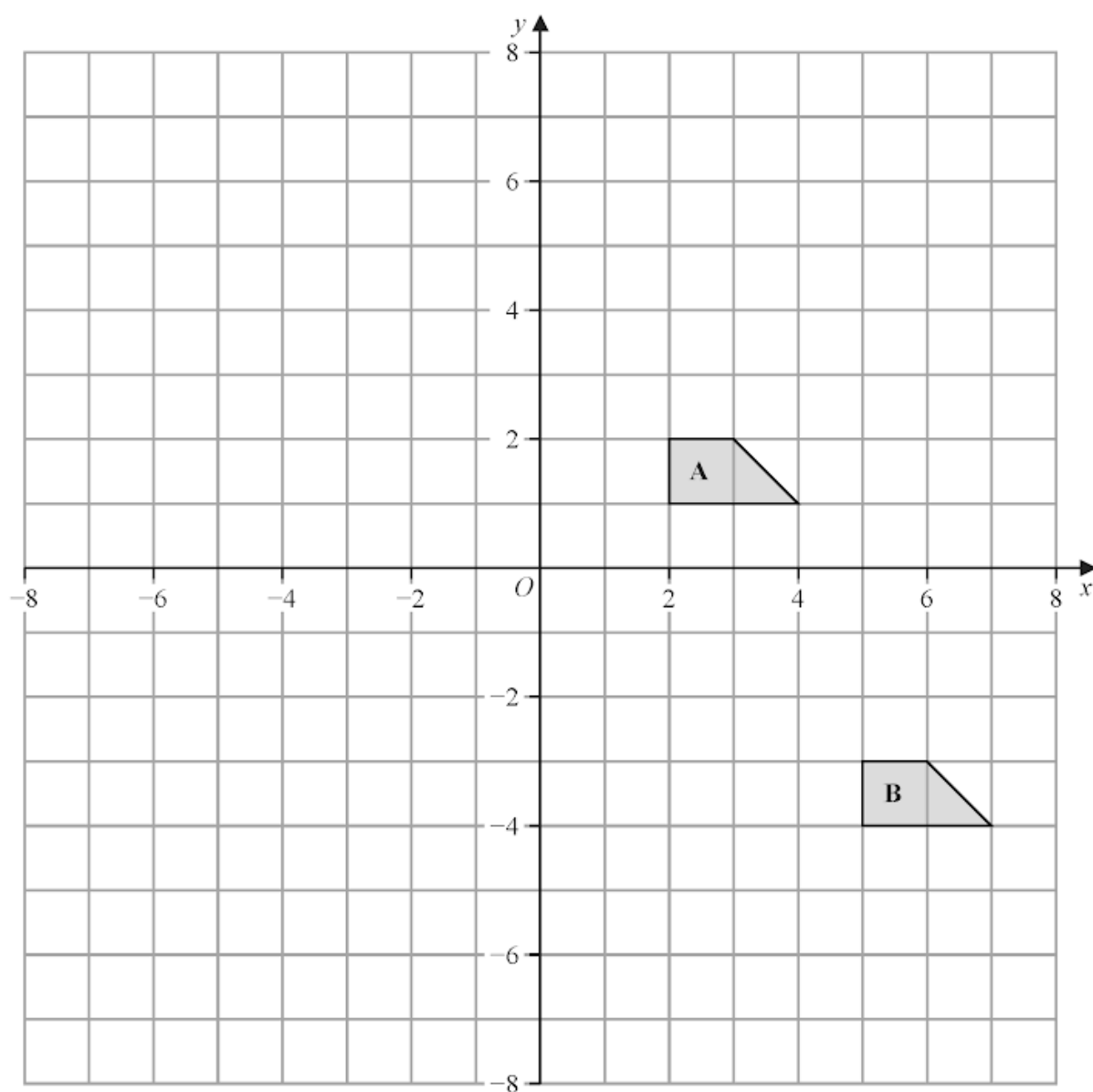
The point A has coordinates $(-5, 2)$

The point B has coordinates $(3, 5)$

Find an equation of the line that passes through B and C

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(4)



- (a) Describe fully the single transformation that maps shape **A** onto shape **B**

(2)

- (b) On the grid above, rotate shape **A** 180° about $(-1, 0)$

Label your shape **C**

(2)

8. 4MA1/2H_Summer_2024_Q24

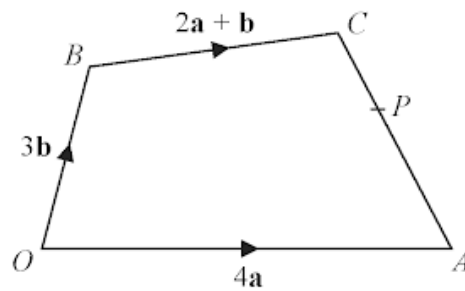


Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

9. 4MA1/2HR_Winter_2023_Q16

10. 4MA1/2HR_Winter_2023_Q2

11. 4MA1/2HR_Summer_2022_Q25

12. 4MA1/2HR_Summer_2022_Q1

13. 4MA1/2H_Summer_2022_Q21

14. 4MA1/2HR_Winter_2021_Q5

15. 4MA1/2HR_Winter_2020_Q26

16. 4MA1/2H_Winter_2020_Q23

17. 4MA1/2HR_Summer_2020_Q25

18. 4MA1/2H_Summer_2020_Q19

19. 4MA1/2H_Winter_2019_Q23

20. 4MA1/2H_Summer_2019_Q14

21. 4MA1/2H_Summer_2019_Q10

22. 4MA1/2HR_Summer_2018_Q23

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